

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	cell membrane	extra ticks negates marks	1	AO1 4.1.1.2
01.2	engulf pathogens	extra ticks negates marks	1	AO1 4.2.2.3 4.3.1.6
	produce antibodies		1	
	produce antitoxins		1	
01.3	2050 – 100	an answer of 1950 scores 2 marks	1	AO2 4.3.1.2
	= 1950	allow 1 mark for a correct subtraction of incorrect values	1	
01.4	any one from: (more) people vaccinated (more) people immune - no new measles strain	ignore injections / treatments / medicines unqualified allow vaccine produced allow (more people given) MMR (vaccine) do not allow antibiotics ignore less people infected	1	AO2 4.3.1.1 4.3.1.2 4.3.1.7
01.5	any one from: - measles is (caused by) a virus - viruses cannot be killed / destroyed by antibiotics	allow measles is not caused by a bacterium allow antibiotics only kill / destroy bacteria ignore harmed / treated	1	AO1 4.3.1.1 4.3.1.8 4.3.1.2
01.6	any one from: - use of a barrier method of contraception - use of a condom - vaccination / immunisation - avoid sexual intercourse / contact	ignore use of diaphragm ignore use protection / safe sex do not accept less sexual intercourse / contact	1	AO1 4.3.1.3

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01.7	any one from: - size / shape/ type of paper disc - concentration of antibiotic - volume / amount of antibiotic (incubation) time (incubation) temperature	ignore paper disc unqualified allow strength / dosage of antibiotic allow 3 days ignore size of petri dish	1	AO2 4.3.1.8
01.8	to check that the disc / water did not have an effect or to make sure it was the antibiotic that had an effect	allow for comparison with the antibiotics allow as a (experimental) control do not accept as a control variable	1	AO3 4.3.1.8
01.9	(antibiotic) A any one from: (antibiotic A) had the largest clear area around it (antibiotic A) killed the most bacteria	no marks if wrong antibiotic given	1 1	AO3 4.3.1.8
Total			13	

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02.1	glucose	extra ticks negates marks	1	AO1 4.4.1.1
	oxygen		1	
02.2	count the number of bubbles produced in 1 minute	extra ticks negates marks	1	AO3 4.4.1.2
	measure the volume of gas produced in 30 seconds		1	
02.3	any one from: - to control the temperature - temperature affects the rate of photosynthesis	ignore reference to 'it' allow so pondweed / solution did not warm up allow correct description of effect of temperature on rate allow high temperatures denature enzymes ignore references to limiting factors	1	AO3 4.4.1.2
02.4	52		1	AO2 4.4.1.2
02.5	all points plotted correctly	(where a bar chart has been plotted) allow 1 mark for all bars plotted correctly if points are plotted as well as bars, ignore bars	2	AO2 4.4.1.2
		allow $\pm \frac{1}{2}$ a square allow 1 mark for three points correctly plotted		
	smooth curve drawn through all points	ignore extensions of line / curve unless inconsistent with line / curve drawn	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	any one from: - the nearer the light source to the pondweed the faster the rate of photosynthesis - the greater the light intensity the faster the rate of photosynthesis	allow converse statements for all marking points allow the nearer the light source to the pondweed the faster the bubbles produced allow the greater the light intensity the faster the bubbles produced allow the closer the light source the more the plant photosynthesises ignore more bubbles are produced with no reference to rate allow oxygen for bubbles do not accept carbon dioxide	1	AO3 4.4.1.2
Total			10	

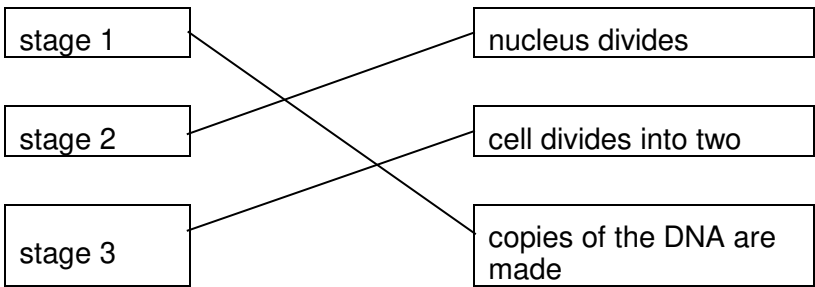
Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	any one from: - water on potato would increase mass - to control amount of water on potato	allow so only the mass of the potato is measured allow to remove water from outside of potato allow liquid / solution / sugar solution for water allow so you get the correct (starting) mass of the potato do not accept so that all the pieces of potato weighed the same	1	AO3 4.1.3.2
03.2	increase in mass increase in length	extra ticks negate marks	1 1	AO2 4.1.3.2
03.3	osmosis into lower	in this order only allow diffusion allow inside do not accept through allow low / more dilute / dilute	1 1 1	AO1 4.1.3.2 AO2 4.1.3.2 AO2 4.1.3.2
03.4	any one from: - the concentration (of sugar solution) in the cells is 0.4 (mol/dm³) - the concentration (of sugar solution) in the cells is the same as the solution (in the tube)	allow reference to potato instead of cells	1	AO3 4.1.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	any two from: • has (root) hairs • large surface / area • (root) hairs extend into soil • (root) hairs have thin walls	ignore references to active transport and mineral uptake allow root hair cells allow wide surface area allow (root) hairs are long / widespread	2	AO1 4.1.1.3 4.1.3.1 4.1.3.2
Total			9	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	breathing rate when walking is twice that at rest	max 2 marks if written in terms of heart rate allow breathing rate when walking is 12 (breaths / minute) more than at rest	1	AO2 4.4.2.2
	breathing rate when jogging is 5 times that at rest	allow breathing rate when jogging is 48 (breaths / minute) more than at rest	1	
	breathing rate when jogging is 2.5 times that when walking	allow breathing rate when jogging is 36 (breaths / minute) more than when walking	1	
		allow for 1 mark if no other marks gained: breathing rate at rest is 12 (breaths per minute), breathing rate when walking is 24 (breaths per minute) and breathing rate when jogging is 60 (breaths per minute) or breathing rate increases with increasing activity		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	(breathing rate increases)	reference to more / faster required at least once for full marks		AO1 4.4.2.2
	to supply more oxygen / O ₂ or to supply oxygen / O ₂ faster	allow to remove more carbon dioxide / CO ₂ or to remove carbon dioxide / CO ₂ faster do not accept incorrectly written formulae	1	
	for (aerobic) respiration or to reduce anaerobic respiration or to reduce lactic acid build up		1	
	(so) that more energy is transferred / released or (because) more energy is required	do not accept used / produced / created or energy made	1	
04.3	right ventricle / side of the heart pumps (blood) to the lungs		1	AO1 4.2.2.2
	left ventricle / side of the heart pumps (blood) to the body		1	
		if no other marks scored allow 1 mark for one side pumps blood to the lungs and the other side pumps blood to the body		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	any one from: • (the left ventricle) has to pump blood further (than the right ventricle) • (the left ventricle) has to pump blood with a greater force (than the right ventricle) • (the left ventricle) has to pump blood at a higher pressure (than the right ventricle)	there must be a comparative statement allow (the left ventricle) has to pump blood all around the body allow (left ventricle) has to pump blood harder	1	AO2 4.2.2.2
04.5	any one from: • strengthens heart (muscle) • reduces chance of another heart attack • reduces / controls weight • improves circulation	ignore prevents / no heart attacks allow decreases chance of fatty deposits or fat building up (in arteries / blood vessels) allow reduces resting heart rate	1	AO2 4.2.2.2 4.2.2.6
Total			10	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	 allow 1 mark for 1 or 2 correct credit can be given where students have matched the boxes correctly, for example numbering the boxes		2	AO1 4.1.2.2
05.2	6 picograms		1	AO2 4.1.2.2
05.3	meristem cells in plants can differentiate throughout the life of the plant		1	AO1 4.1.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	any two from: - may cure / treat diseases or cure medical conditions or produce replacement cells / tissues / organs - cells unlikely to be rejected by patient - cells / tissues of any type can be produced - many cells produced - cells produced could be used for research - would reduce waiting time for transplants	ignore references to cost ignore all reference to producing babies / IVF allow example eg diabetes / paralysis allow cells can be stored for future use ignore used in medical treatments ignore patient makes / grows cells / tissues / organs ignore same genetic information ignore differentiated into most types of cells	2	AO3 4.1.2.3 4.1.1.4 4.6.2.4
05.5	any two from: (potential) life is killed / destroyed - shortage of donors / eggs - egg donation / collection has risks - do not yet know risks / side effects of the procedure on the patient - may transfer (viral) infection - poor success rate	ignore references to cost ignore unethical unqualified ignore reference to religion / beliefs allow embryo is killed ignore embryo is destroyed ignore embryo is a life / becomes a baby ignore long term effects are not well understood allow may cause tumours / cancer allow in terms of viable egg / embryo / cell / tissue / organ production	2	AO3 4.1.2.3 4.1.1.4 4.6.2.4
Total			8	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	nucleus permanent vacuole plasmids prokaryotic cells plant cells only eukaryotic cells allow 1 mark for one or two correct links		2	AO1 4.1.1.1 4.1.1.2
06.2	vacuole ribosome cell wall tick box takes precedence if no tick is given, look at both the figure and the circling of words in the table if writing is seen on the figure and in the table both must be correct		1	AO1 4.1.1.2
06.3	turn the (fine focusing) knob until the cells are in focus	allow focus it do not accept increase magnification ignore decrease magnification ignore clear ignore references to resolution / illumination ignore zoom in / out	1	AO2 4.1.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	(rotate the) nosepiece / objective lens	allow change the (objective / eyepiece) lens	1	AO2 4.1.1.2
	to a higher power (lens)	allow (to) increase the magnification a comparator is required ignore change / adjust the magnification allow stronger or more powerful lens ignore references to resolution / illumination unqualified ignore zoom in / out ignore references to an electron microscope	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	conversion of units: (112 mm →) 112 000 (μm) or (280 μm →) 0.28 (mm)	an answer of 400 (×) scores 3 marks	1	AO2 4.1.1.2
	(magnification =) $\frac{112}{0.28}$ or (magnification =) $\frac{112\,000}{280}$	allow 1 mark for no conversion of units 112 / 280 or incorrect value from step 1 correctly substituted	1	
	400 (×)	do not accept if units are given if no other mark scored allow 1 mark for: magnification = $\frac{\text{size of image}}{\text{size of real object}}$ a triangle with words or letters in is insufficient, as the correct rearrangement is needed	1	
Total			9	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	is not caused by a pathogen / infective organism	allow not caused by a microorganism / microbe ignore not caused by infection ignore named pathogen unless bacteria, virus and fungus all mentioned	1	AO1 4.2.2.4 4.2.2.5 4.3.1.1 4.3
	(so) is not passed / spread (from person to person)	allow cannot be spread / caught allow is not infectious / contagious	1	AO2 4.2.2.4 4.2.2.5 4.3.1.1 4.3
07.2	reduced / restricted / stopped blood flow	allow 'it' for heart it does not matter where blood flow is restricted to – heart / body	1	AO1 4.2.2.4
	(so) less oxygen reaches heart (muscle / cells)	must reference heart / it allow no oxygen reaches the heart (muscle / cells)	1	
	(so heart muscle / cells) cannot respire (enough) or (so heart muscle / cells) do not release (enough) energy	do not accept do not make / produce / create energy ignore references to breathing / suffocation ignore blood clots / blockages	1	

Question	Answers	Mark	AO / Spec. Ref.
02.3	Level 3: Relevant points (factors / effects) are identified, given in detail and logically linked to form a clear account.	5–6	AO2
	Level 2: Relevant points (factors / effects) are identified and there are attempts at logical linking. The resulting account is not fully clear.	3–4	AO2 AO1
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	1–2	AO1
	No relevant content	0	
	Indicative content medical risk factors: • high blood pressure • high cholesterol • diabetes • genetic factors • medications lifestyle risk factors: • smoking • obesity • lack of exercise • high fat / energy diet • eating insufficient fruit / vegetables • alcohol • high salt intake • exposure to air pollution • certain drugs / correct named drug examples of links: • smoking – high bp / cholesterol / fatty deposition • obesity – lack of exercise / high bp / cholesterol / fatty deposition / diabetes • exercise – obesity / bp / diabetes • diet – obesity / cholesterol / diabetes • alcohol – bp / cholesterol • high salt intake - high blood pressure • genetic factors – bp / cholesterol / diabetes / obesity • medication – can affect blood / blood vessels / metabolism the main discriminator is the quality of linking both lifestyle and medical factors are required for level 3		4.2.2.2 4.2.2.4 4.2.2.5 4.2.2.6
Total		11	